

Heat Transfer to a Fluid in a Stirred Tank

Pre-Lab Plan

Unit Operations Lab # 1

Date: 01/21/2020

Commented [DL1]: 31/35

Group 4 - Tuesday PM section

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1. Experimental Objective (5 pts)

The objective of this experiment is to calculate the overall heat transfer coefficient, U , (HTC) range for a variety of heat transfer processes. These studies are done by measuring HTC between the fluid and the interior of the vessel and a correlation will be drawn of its dependence on speed of the impeller, baffles and properties of the fluid. For these experiments, the independent variables are the agitation rate of the impeller in RPM and the heat exchanger flow rate in GPM, while the dependent variables are the condensate temperature and the heat transfer coefficient.

Commented [DL2]: 4.5/5

Commented [AZ3]: Unsteady or steady state?

Commented [AZ4]: Good summary of the objective

2. Experimental

The apparatus and the materials and supply of this experiment are introduced below and briefly explained their usage.

2.1 Apparatus (5 pts)

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A brief introduction of the apparatus is needed. You may include the flowsheet of the process.

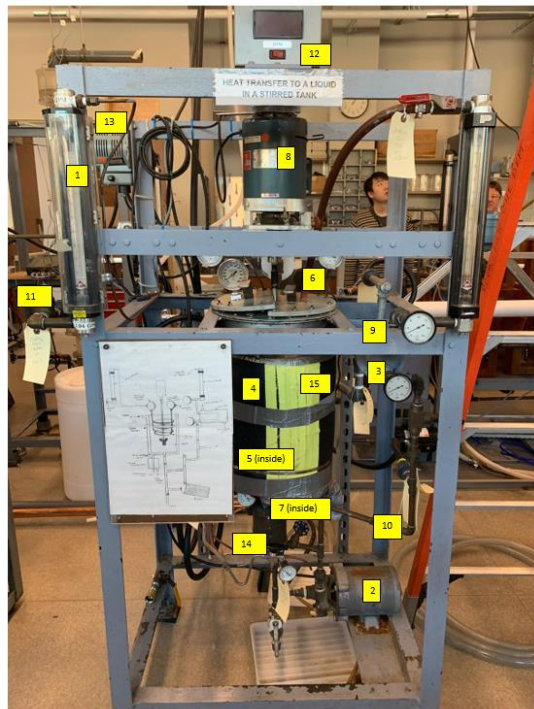


Figure 1. Heat Transfer System

Table 1: Apparatus used in Heat Transfer to a Fluid in a Stirred Tank experiment

Serial No.	Apparatus	Description
1.	flowmeter (2)	Device to measure the fluid flowrate inside the system
2.	Pump	Mechanical device use to push fluid through the piping system
3.	Cooler Tank	Jacketed vessel with recirculation of steam and water through a cooling coil
4.	Steam Jacket	Enclosure surrounding the tank, which allows the steam to flow
5.	Cooling Coil	Coil tubing inside the tank, which allows the cold water to flow
6.	Thermometers	Device that measures the temperature of the flowing fluid at different locations inside the unit.
7.	Impeller	Agitator that stirs the liquid inside the tank, ensures a uniform mixing. It can be adjusted for higher speeds
8.	Motor	Mechanical device that drives the impeller inside the tank by supplying power
9.	Heat Exchanger	Shell and tube heat exchanger type, which allows the transfer of heat between a cold and hot stream
10.	Piping	Tubing system which allows fluid to be transported in the unit
11.	Valves	Device that controls the fluid flow inside the piping
12.	Tachometer	Device use to measure the RPM
13.	Variac	Device use to set the voltage delivered to the system
14.	Stream Trap	Use to contain the steam coming out of the steam jacket
15.	Insulator	Material surrounding the tank sue to minimize heat loss or gain from the surrounding

Commented [DL6]: Include all gauges and valves.**Commented [DL7]:** Unit and the maximum capacity.**Commented [AZ8]:** A better name would be 'constant stirred tank, (or vessel)'**Commented [AZ9]:** Insulation**2.2 Materials and Supplies (5 pts)****Table 2:** Materials used in Heat Transfer to a Fluid in a Stirred Tank experiment

SNo.	Material	Description
1.	Measuring scale	For measuring the size of the tank
2.	Water	Water used to fill the tank and used as cooling water in the heat exchanger and condensate steam
3.	Steam	Used as a hot stream of steam through the steam jacket to heat the water in the unit
4.	Baffles	Used to promote mixing by creating turbulence and increase heat transfer in the tank
5.	Stopwatch	Device use to record time during collection of fluid or condensate
6.	Graduate Cylinder	Device used to measure the volume of fluid collected

Commented [DL10]: 5/5**Commented [AZ11]:** Serial no.**Commented [DL12]:** Why?

2.3 Experimental Plan (10 pts)

- General Operating Procedure

1. Fill the tank with water until the level of liquid is about 2 inches from the top of the container.
2. Turn on impeller and adjust its speed to 7600 RPM.
3. Turn on cooling water to the cooling coils. Fully open the control valves.
4. Turn on the cooling water to the tube side of the shell and to the tube heat exchanger.
5. Verify that the outlet valve for the discharge of the pump is completely closed, also the suction and bypass valves are completely open.
6. Turn on the pump and let it run for a couple seconds with the discharge valve closed.
7. Slowly open the discharge valve, followed by setting the recirculation rate on the bypass valve to one that reaches steady state. Ensure that the circulating liquid is fed back to the tank.
8. Turn on the cooling water to the steam condenser, follow by slowly open the inlet steam valve around 1 ½ turns.
9. Ensure that all the steam condenses completely in the jacket by verifying a small temperature increase in the shell side fluid of the steam condenser.
10. Record all the data after steady state has been reached.

- Experiment A: Wall HTC for the Unbaffled Agitated Tank

1. Set a constant steam flow rate to the vessel for a impeller speed of approximate 600 RPM, while water is flowing through the cooling coils, and the heat exchanger flowrate in order to keep the temperature inside the vessel at 65°C.
2. Record data after steady state has been reached.
3. Repeat first two steps for higher impeller speed, such as 800 and 1000 RPM.
4. Repeat first two steps at 800 RPM and adjust the heat exchanger to two different flowrates, recording two different vessel temperatures.

- Experiment B: Wall HTC for the Baffled Agitated Tank

1. Set a constant steam flow rate to the vessel for a impeller speed of approximate 600 RPM, while water is flowing through the cooling coils, and the heat exchanger flowrate in order to keep the temperature inside the vessel at 65°C.
2. After steady state has been reached, gently lift the vessel cover plate and slip the baffles.
3. Ensure that top flat section is seated over several allen screw heads.
4. After steady state has been reached, record all the data.
5. Repeat above steps for 800 and 1000 RPM.
6. Repeat the procedure at 800 RPM and adjust the heat exchanger to two different flowrates, recording two different vessel temperatures.

- Experiment C: Unsteady State HTC for the Unbaffled Agitated Tank

1. Start with 800 RPM speed for the impeller, and with a temperature lower than 40°C, open the steam valve a few turns.
2. Record the fluid temperature as a function of time until 85°C.
3. Turn off the steam and recirculate water through the heat exchanger and cooling coils so the fluid in the vessel can be cooled.
4. Repeat the procedure using another speed for the impeller, such as 1000 RPM

Commented [DL13]: 8/10

Commented [DL14]: Too high. Less than 1000 rpm.

Commented [AZ15]: Use trial and error to come up with impeller speeds that provide a good range but don't cause too much splashing at the top of the vessel.

Commented [DL16]: Be specific for all rates in this section.

Commented [DL17]: What's the criteria of the steady state?

Commented [DL18]: List all parameters need to be measured.

Commented [DL19]: Choose 2 different experiments among those 3 listed.

Project Safety Evaluation

Project Title: *Heat Transfer to a Fluid in a Stirred Tank*

Date: 01/21/2020

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Agitators and electric pump next to the water source
Potential mechanical hazards?	Yes	If yes, identify conditions: Movement of propeller that will be used for agitation
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	Yes	If yes, identify conditions: Burn hazard due to steam and heated pipes
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	Yes	If yes, specify type: Thermal gloves to handle the hot equipment whilst steam flow
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Water
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	85	Due to steam
Pressure (psia)	14.7	Atmospheric pressure. System is not pressurized.
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
	Yes	Power shut off switches and control valves

Commented [DL20]: 9.5/10

Commented [AZ21]: What is the hazard?

Commented [DL22]: CAS# required.

Commented [DL23]: 120 degree Celsius.

Commented [AZ24]: Cooling water is pressurized; tap water is normally around 20psig.

Commented [AZ25]: Valves for what? (Steam, cooling water)

I have read relevant background material for the Unit Operations Laboratory entitled: *"Heat Transfer to a Fluid in a Stirred Tank"* and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

signature of team member: Jasmine Villegas 01/21/2020

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signature of team member: Nisha Patel 01/21/2020

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